

Nirbhay Sharma

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Education

M.Tech, AI | Indian Institute of Science (IISc) Bangalore | **AIR-6** GATE DA | CGPA: **9.3/10** 08/24-05/26
B.Tech, CSE | Indian Institute of Technology (IIT) Jodhpur | CGPA: **8.97/10** 08/19-05/23

Technical Skills

Programming Languages: Python, C/C++ | **Skills:** Machine Learning, Deep Learning, Computer Vision
Tools and Frameworks: Pytorch, Django, Flask, Docker, AWS Lambda, Regex, Git, Github, Firebase, MySql

Publications

- **An Extremely Lightweight CNN Model For the Diagnosis of Chest Radiographs in Resource-constrained Environments** | *International Journal of Medical Physics* 2023 | [Paper](#)
- **Aggregation-Assisted Proxyless Distillation: A Novel Approach for Handling System Heterogeneity in Federated Learning** | *International Joint Conference on Neural Networks (IJCNN)* 2024 | [Paper](#)

Industry Experience

Mastercard India | **Intern** (Data Scientist) 05/25-07/25

- Worked with **AI Garage** on **Multi-Teacher GNN to MLP** knowledge distillation for efficient inference
- Trained multiple teacher **GNN** models like **GCN**, **GAT**, **SAGE** on **node and edge prediction**, **contrastive** tasks
- Proposed **GMOE2MLP**, a **graph mixture of expert to MLP** distillation method and a novel **cluster contrast loss** to distill the **embeddings from best teacher to MLP**

Faaya Astu India | **Full Time** (ML Engineer) 06/23-06/24

- Trained **Stable Diffusion ControlNet** models on **Lineart** and **Colorbox** control on **VastAI** GPU instance and deployed them on **RunPod** for more flexibility and control on print generation
- Trained **Low Rank Adaptation (LoRA)** models using **Kohya_SS** for custom **face** and **background** generation
- Experimented with custom **ComfyUI** workflows with integrated **ControlNet**, **LoRA**, **InstantID** models
- Containerised **ComfyUI** with **Docker** and deployed them as **Serverless Endpoints** on **RunPod** and exposed endpoint APIs to **AWS Lambda** to create **APIs for APP** using **AWS API gateway**

ExaWizards India | **Intern** (AI Engineer) 06/22-07/22

- Worked on **Split Neural Network** ML paradigm and Splitted **Mask-RCNN**, **FCN_Resnet50**, **YOLOv5** models for **Instance segmentation**, **segmentation**, **face detection** tasks
- Implemented **Autoencoder** model for efficient **image compression to latent space** and setup **Pysyft** to communicate latents from **Jetson Nano** to GPU server, **preserving data privacy** at Jetson Nano

Projects

Leveraging Generative Modelling for Self Supervised Learning | [Github](#) | [Report](#)

- Leveraged **Neural ODE** to model the representation space as **dynamical system**
- Utilized **Energy Based Models (EBMs)** and proposed **LEMa** to elevate representation quality via genAI
- Observed a boost of **3-6%** on **linear evaluation** accuracy on **CIFAR10/100** with **Resnet18/50** encoders

Self Supervised Representation Learning Algorithms (SSL) | [Github](#)

- Performed a comparative study on **6** SSL methods like **SimCLR**, **BYOL**, **SupCon** etc. from **scratch** in pytorch
- Proposed **DARe**, a **distribution alignment regularizer** for learning rich representations
- Observed a boost of **2-5%** on **linear evaluation** accuracy on **CIFAR10/100** with **Resnet18/50** encoders

Regularizing Federated Learning (FL) via Adversarial Model Perturbations (AMP) | [Github](#)

- Analyzed the effect of integrating **AMP** on SOTA FL algorithms like **FedAvg**, **FedProx**, **FedNTD**, **SCAFFOLD**. Observed a boost of **2-3%** accuracy on **CIFAR10/100** dataset with **AMP** integrated FL algorithms

Image Captioning using Detection Transformer (DeTR) | [Github](#)

- Implemented modified **DeTR** from **scratch** in **pytorch** for **image captioning** task. Trained **DeTR** on **Flickr30k** dataset for **500** epochs and achieved a **BLEU** score of **57.36** on **Flickr8k** dataset